

```
def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    return n * factorial(n - 1)  
  
n = int(input("Enter n: "))  
  
print("Factorial =", factorial(n))  
print("Recursive Time Complexity = O(n)")
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents/Big-O, Ω and Θ.py ====  
Enter n: 5  
Factorial = 120  
Recursive Time Complexity = O(n)  
>>>
```